was similar to that of de Broglie, in which particles were guided by a "quantum potential" constructed from the wave function. Today this approach is often known as the *de Broglie–Bohm theory*, or simply *Bohmian mechanics*. Bohm's presentation of the theory was a bit more fleshed out than de Broglie's, especially when it came to describing the measurement process.

Even today you will sometimes hear professional physicists say that it's impossible to construct a hidden-variable theory that reproduces the predictions of quantum mechanics, "because of Bell's theorem." But that's exactly what Bohm did, at least for the case of non-relativistic particles. John Bell, in fact, was one of the few physicists who was extremely impressed by Bohm's work, and he was inspired to develop his theorem precisely to understand how to reconcile the existence of Bohmian mechanics with the purported no-hidden-variables theorem of von Neumann.

What Bell's theorem actually proves is the impossibility of reproducing quantum mechanics via a *local* hidden-variables theory. Such a theory is what Einstein had long been hoping for: a model that would attach independent reality to physical quantities associated with specific locations in space, with effects between them propagating at or below the speed of light. Bohmian mechanics is perfectly deterministic, but it is resolutely nonlocal. Separated particles can affect each other instantaneously.

Bohmian mechanics posits both a set of particles with definite (but unknown to us, until they are observed) positions, and a separate wave function. The wave function evolves exactly according to the Schrödinger equation—it doesn't even seem to recognize that the particles are there, and is unaffected by what they are doing. The particles, meanwhile, are pushed around according to a guidance equation that depends on the wave function. However, the way in which any one particle is guided depends not just on the wave function but also on the positions of *all the other particles* that may be in the system. That's the nonlocality; the motion of a particle here can depend, in principle, on the positions of other particles arbitrarily far away. As Bell himself later put it, in Bohmian mechanics "the Einstein-Podolsky-Rosen paradox is resolved in the way which Einstein would have liked least."

This nonlocality plays a crucial role in understanding how Bohmian mechanics reproduces the predictions of ordinary quantum mechanics.